Winter 2016, MATH 215 Calculus III, Exam 1

2/11/2016, 6:10-7:40pm (90 minutes)

1. (5 points) Find an equation of the sphere centered at the point $(4, 1, 2)$ and is tangential to the plane $x - 2y + 2z = 3$.

2. (5 points) Let $f(x, y)$ be a differentiable function. Suppose that the rate of change of f at the point $P(1, 2)$ in the direction from P to $Q(0, 1)$ is $\sqrt{2}$. Suppose furthermore that the directional derivative $D_{\vec{u}}f(1, 2) = -1$ where $\vec{u} = \frac{1}{2}\vec{i} - \frac{\sqrt{3}}{2}\vec{j}$. Find the partial derivatives f_x and f_y at $P(1, 2)$.

3. (5 points) Consider the function $f(x, y, z) = x^2 + y^2/2 + 2z^2 + 2xz$. Find all points on the level surface $f(x, y, z) = 4$ at which the tangent plane is parallel to the xy-plane.

4. (5 points. No partial points) Find a parametric equation of the curve of the intersection of the paraboloid $z = 4x^2 + y^2$ and the cylinder $x = y^2$.

5. (5 points) Consider the vector $\vec{v} = \langle 3, 1, 5\rangle$. Is it possible to decompose this vector into the form $\vec{v} = \vec{A} + \vec{B}$ where the vector $\vec{A}$ is parallel to the line $\vec{r}(t) = \langle -3, -1, -4\rangle + t\langle 1, 2, 3\rangle$ and the vector $\vec{B}$ is perpendicular to the plane $2x + y + 2z = -10$, respectively? If your answer is yes, then find $\vec{A}$ and $\vec{B}$. If your answer is no, explain why in detail.

6. (5 points) Consider the vector $\vec{v} = \langle 3, 1, 5\rangle$. Is it possible to decompose this vector into the form $\vec{v} = \vec{A} + \vec{B}$ where the vector $\vec{A}$ is parallel to the plane $x + y + z = 2016$ and $\vec{B}$ is perpendicular to the same plane? I If your answer is yes, then find $\vec{A}$ and $\vec{B}$. If your answer is no, explain why in detail.

7. (5 points) Find the length of the curve $\vec{r}(t) = \langle 2t, t^2, \frac{1}{3}t^3\rangle$, $0 \leq t \leq 1$.

8. (5 points. No partial points) Which of the following equations does the function $z = f(x+t)+g(x-t)$ satisfy for all differentiable functions $f(s)$ and $g(s)$ in a single variable?

   (a) $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial t} = 0$

   (b) $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial t}$

   (c) $\frac{\partial z}{\partial x} + \frac{\partial^2 z}{\partial t^2} = 0$

   (d) $\frac{\partial z}{\partial x} = \frac{\partial^2 z}{\partial t^2}$

   (e) $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial t^2} = 0$

   (f) $\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial t^2}$

9. (5 points) Consider the two lines $\vec{r_1}(t) = \langle 1, 1, 0\rangle + t\langle 1, -1, 2\rangle$ and $\vec{r_2}(t) = \langle 2, 0, 2\rangle + t\langle -1, 1, 0\rangle$. Is there a plane which contains both of these lines? If your answer is yes, then find an equation of the plane. If your answer is no, explain the reason in detail.

10. (5 points) Consider the two lines $\vec{r_1}(t) = t\langle 1, 1, 1\rangle$ and $\vec{r_2}(t) = \langle -1, 0, 0\rangle + t\langle 1, 2, 3\rangle$. Is there a plane which contains both of these lines? If your answer is yes, then find an equation of the plane. If your answer is no, explain the reason in detail.